

Dr. Gyu Seon Kim

Last update on October 1, 2026

Aerospace AI Research Engineer

R&D Center, RealtimeVisual Co., Ltd., Seoul, Republic of Korea

Member of the IEEE

- Email: kingdom0545@korea.ac.kr
- Personal Website: <https://gyuseonkim.github.io/>
- Google Scholar: <https://scholar.google.com/citations?user=530QU2AAAAAJ>
- LinkedIn: www.linkedin.com/in/gyu-seon-kim-67b849391
- DBLP: <https://dblp.org/pid/333/1068.html>
- DBPia (Korean): <https://www.dbpia.co.kr/author/authorDetail?ancId=5405761>
- GitHub: <https://github.com/GSKIMKOR>
- IEEE Member: <https://ieeexplore.ieee.org/author/37089737473>

Research Interests

- **Autonomous Aerospace Systems:** Control and Communication for Autonomous Aerospace Systems (*e.g.*, Aircraft, Drone, UAV, UAM, Fixed Wing, Rotary Wing, Rotorcraft, Launch Vehicles, and Spacecraft)
- **Physical AI for Reinforcement-Learning-based Flight Control:** Developing reinforcement-learning-based flight controllers and enabling their closed-loop sim-to-real deployment on fighter aircraft and other real-world aerial platforms.
- **AI for Defense and Military Applications:** Intelligent Control and Autonomous Decision-Making for Defense and Aerospace Systems (*e.g.*, Fighter Aircraft, Military Drone, Anti Drone System, Missile, Decoy, and Torpedo)
- **MARL-based Autonomous Control:** Multi-Agent Reinforcement Learning for Cooperative and Networked Aerial Vehicle Systems
- **RL-based Satellite Communication Systems:** Reinforcement Learning-based Communication and Networking for Satellite Systems (*e.g.*, LEO Satellite Routing, Cube Satellite Design, Satellite Constellations, and Space-Air-Ground Integrated Networks)
- **Quantum AI:** Quantum Reinforcement Learning for Aerospace Control and Communication Systems

Educational Background

- **Korea University – College of Engineering**, Seoul, Republic of Korea
– Ph.D. in **Electrical and Computer Engineering (ECE)**
- **Inha University – College of Engineering**, Incheon, Republic of Korea
– B.S. in **Aerospace Engineering**

Academic Background

- **Korea University – Research Institute of Information and Communication Technology**, Seoul, Republic of Korea
– *Postdoctoral Researcher*
- **University of Houston – College of Engineering (ECE)**, Houston, TX, USA
– *Visiting Scholar*, Wireless Networking, Signal Processing and Security Laboratory (Prof. Zhu Han)
- **Korea University – College of Engineering (ECE)**, Seoul, Republic of Korea
– *Research Assistant*, Artificial Intelligence and Mobility Laboratory (Advisor: Prof. Joongheon Kim)

Industry Background

- **RealtimeVisual – R&D Center**, Seoul, Republic of Korea
– *Aerospace AI Research Engineer (Military Service Exception)*

Awards and Honors

- **IEIE Best Paper Award** (11/2025)
– Title: *Lightweight Control Modeling via Quantum Reinforcement Learning*
 - **IPESK Next Generation Engineering Researcher Award** (10/2025)
– Funded by an Institute for Promotion of Engineering and Science of Korea
 - **IEEE Vehicular Technology Society Student Scholarship Award** (10/2025)
– Funded by an IEEE VTS Endowment Fund and partially funded by a VTS Award Fund, and administered by the IEEE Foundation
 - **KICS Best Paper Award** (11/2024)
– Title: *Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats*
 - **IEEE Seoul Section Student Paper Contest, Best Paper Award** (12/2023)
– Title: *Aircraft Taxi Routing using Reinforcement Learning at Hartsfield Jackson Atlanta International Airport*
 - **KICS Best Paper Award** (02/2023)
– Title: *Self-Learning-Based Hybrid MAC for Military UAV Networks*
-

R&D Projects

- **18. Development of an Air-to-Air Combat Software Verification Environment**
Funded by ADD (Agency for Defense Development)
- **17. KE-Anduril AI Pilot Project: Development of AI Algorithms and Simulation Frameworks for Mission Autonomy**
Funded by Korean Air
- **16. Discovering Quantum-Advantage Cases for Industry Adoption**
Funded by KTL (Korea Testing Laboratory) and the Ministry of Trade, Industry and Resources (MOTIR), Republic of Korea
- **15. LEO Satellite Routing Research using Large Language Model and Reinforcement Learning**
Funded by ETRI (Electronics and Telecommunications Research Institute)
- **14. Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks**
Funded by ETRI (Electronics and Telecommunications Research Institute)
- **13. Research on Quantum Multi-Agent Reinforcement Learning Stability (Research on Multi-Agent Reinforcement Learning Exploration, Communication, Training Strategy)**
Funded by ETRI (Electronics and Telecommunications Research Institute)
- **12. Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center, ITRC**
Funded by IITP (Institute of ICT Promotion)
- **11. Quantum AI Empowered Second-Life Platform Technology**
Funded by IITP (Institute of ICT Promotion), Software Technology Advanced Research
- **10. Development of a 20-Ton Excavator and Application Services using the SDM (Software-Defined-Machine) Platform**
Funded by KOCETI (Korea Construction Equipment Technology Institute) and Hyundai
- **9. Research on Learning-based Swarm Mission Planning Algorithms**
Funded by LIG Defense & Aerospace and ADD (Agency for Defense Development)
- **8. AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development**
Funded by NRF (National Research Foundation of Korea)
- **7. Quantum Hyper-Driving: Quantum Hyper-Connected and Hyper-Sensing Autonomous Mobility**
Funded by NRF (National Research Foundation of Korea)
- **6. Distributed Learning System Design and Implementation for Clinical Applications**
Funded by CIPHEROME
- **5. Routing Algorithms for LEO Satellite Networks**
Funded by Solvit System
- **4. Optimal Positioning Algorithms for Wide-Area Relaying Networks**
Funded by Solvit System
- **3. Advancement Technology Development for Submarine Target Identification and Engagement Support Intelligence**
Funded by LIG Defense & Aerospace
- **2. Advancement Technology Development for Torpedo Deception Strategies in Submarines**
Funded by LIG Defense & Aerospace
- **1. Nano UAV Intelligence Systems Research Lab (NUISRL)**
Funded by ADD (Agency for Defense Development)

Publications (International)

Journals (SCI) and Magazines

◀ Review ▶

[TIV.review] [28] **Gyu Seon Kim**, Hyeon Lee, Soohyun Park, Soyi Jung, Zhu Han, and Joongheon Kim, "LLM-Guided Reinforcement Learning for Carbon-Neutral Intelligent Vehicle Control," *IEEE Transactions on Intelligent Vehicles*, (Under Review).

[TMC.review] [27] **Gyu Seon Kim**, Hyojun Ahn, Zhu Han, and Joongheon Kim, "Quantum Reinforcement Learning for Carbon-Aware Energy-Efficient Air Traffic Control," *IEEE Transactions on Mobile Computing*, (Under Review).

◀ Revision ▶

[TASE.revision] [26] **Gyu Seon Kim** and Joongheon Kim, "(Double-Blind Review)," *IEEE Transactions on Automation Science and Engineering*, (Under Revision).

[IoTJ.revision] [25] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "LLM-Guided Deep Reinforcement Learning for Robust and Autonomous Anti-Drone Systems," *IEEE Internet of Things Journal*, (Under Revision).

[TSC.revision] [24] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "PID-Based Sun-Tracking Attitude Control for a Laboratory CubeSat and Model-Estimated Solar Charging," *IEEE Transactions on Sustainable Computing*, (Under Revision).

[TMC.revision] [23] **Gyu Seon Kim**, Jaehyun Chung, Soyi Jung, Soohyun Park, Trung Q. Duong, and Joongheon Kim, "Joint Mobility Control and Networking in Autonomous Aircraft Cooperation: A Quantum Reinforcement Learning Approach," *IEEE Transactions on Mobile Computing*, (Under Revision).

◀ Published ▶

- [TMC.accept] [22] **Gyu Seon Kim**, Emily Jimin Roh, Soyi Jung, Soohyun Park, and Joongheon Kim, "Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments," *IEEE Transactions on Mobile Computing*, 2026 (Early Access).
- [Netw.accept] [21] Hyunsoo Lee, **Gyu Seon Kim**, Emily Jimin Roh, Soyi Jung, Soohyun Park, Trung Q. Duong, and J. Kim, "Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks," *IEEE Network*, 2026 (Early Access).
- [TIV.accept] [20] **Gyu Seon Kim**, Sungjoon Lee, Taejin Woo, and Soohyun Park, "Cooperative Reinforcement Learning for Military Drones over Large-Scale Battlefields," *IEEE Transactions on Intelligent Vehicles*, 2024 (Early Access).

◀ 2026 ▶

- [CM'26.10] [19] **Gyu Seon Kim**, Jaehyun Chung, Soohyun Park, Soyi Jung, Trung Q. Duong, and Joongheon Kim, "Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective," *IEEE Communications Magazine*, vol. 64, no. 10, pp. 46-53, October 2026.
- [IoTJ.26.09] [18] **Gyu Seon Kim**, Emily Jimin Roh, Soohyun Park, and Joongheon Kim, "Exponentially Scalable Quantum Reinforcement Learning for Large-Scale Network Scheduling," *IEEE Internet of Things Journal*, vol. 13, no. 18, pp. 43910-43925, September 2026.
- [ToN.26.08] [17] Hyojun Ahn, Seungcheol Oh, **Gyu Seon Kim**, Soohyun Park, Soyi Jung, and Joongheon Kim, "Hallucination-Aware Hierarchical LLM for Autonomous UAV Mobility Control: A Safe Reinforcement Learning Approach," *IEEE Transactions on Networking*, vol. 34, pp. 6827-6842, August 2026.
- [IoTJ.26.08] [16] **Gyu Seon Kim**, Chaemoon Im, Yeong Goo Kim, Jaekyoung Ha, and Joongheon Kim, "Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks," *IEEE Internet of Things Journal*, vol. 13, no. 16, pp. 37402-37414, August 2026.
- [TMC.26.07] [15] Soohyun Park, **Gyu Seon Kim**, Soyi Jung, Zhu Han, and Joongheon Kim, "Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient CubeSats," *IEEE Transactions on Mobile Computing*, vol. 25, no. 7, pp. 10385-10401, July 2026.
- [TAES.26.03] [14] **Gyu Seon Kim**, Jaehyun Chung, Soyi Jung, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 3838-3863, March 2026.
- [TAES.26.02] [13] **Gyu Seon Kim**, Soohyun Park, Soyi Jung, David Mohaisen, and Joongheon Kim, "Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 682-703, February 2026.
- [IoTJ.26.02] [12] Hyojun Ahn, **Gyu Seon Kim**, In-Sop Cho, Soyi Jung, and Joongheon Kim, "Hybrid Large Language Models and Reinforcement Learning for Energy-Efficient Multi-Satellite Scheduling: Boosting the Performance from Scratch," *IEEE Internet of Things Journal*, vol. 13, no. 3, pp. 5379-5392, February 2026.
- [TMC.26.01] [11] **Gyu Seon Kim**, Yeryeong Cho, Jaehyun Chung, Soohyun Park, Soyi Jung, Zhu Han, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks," *IEEE Transactions on Mobile Computing*, vol. 25, no. 1, pp. 1200-1218, January 2026.

◀ 2025 ▶

- [TON.25.08] [10] Hankyul Baek[†], **Gyu Seon Kim (co-first)**[†], Soohyun Park, Andreas F. Molisch, and Joongheon Kim, "Slimmable Federated Reinforcement Learning for Energy-Efficient Proactive Caching," *IEEE Transactions on Networking*, vol. 33, no. 4, pp. 2079-2094, August 2025.
- [†]These authors contributed equally.
- [TAES.25.08] [9] **Gyu Seon Kim**, Yeryeong Cho, Soohyun Park, Soyi Jung, and Joongheon Kim, "Quantum Multiagent Reinforcement Learning for Joint Cube Satellites and High-Altitude Long-Endurance Aerial Vehicles in SAGIN," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 4, pp. 9490-9510, August 2025.
- [IoTJ.25.07] [8] **Gyu Seon Kim**, Sungjoon Lee, In-Sop Cho, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Lightweight LEO Satellite Routing," *IEEE Internet of Things Journal*, vol. 12, no. 14, pp. 28986-29004, July 2025.
- [IoTJ.25.07] [7] Soohyun Park, **Gyu Seon Kim**, and Joongheon Kim, "Joint Quantum Reinforcement Learning and Neural Myerson Auction for High-Quality Digital-Twin Services in Multi-Tier Networks," *IEEE Internet of Things Journal*, vol. 12, no. 13, pp. 23722-23735, July 2025.

◀ 2024 ▶

- [IoTJ'24.12] [6] Soohyun Park[†], **Gyu Seon Kim (co-first)**[†], Soyi Jung, and Joongheon Kim, "Markov Decision Policies for Distributed Angular Routing in LEO Mobile Satellite Constellation Networks," *IEEE Internet of Things Journal*, vol. 11, no. 23, pp. 38744-38754, December 2024.
- [†]These authors contributed equally.
- [CM'24.10] [5] Soohyun Park, **Gyu Seon Kim**, Zhu Han, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning is All You Need: Coordinated Global Access in Integrated TN/NTN Cube-Satellite Networks," *IEEE Communications Magazine*, vol. 42, no. 10, pp. 86-92, October 2024.
- [TVT'24.08] [4] **Gyu Seon Kim**, Jaehyun Chung, and Soohyun Park, "Realizing Stabilized Landing for Computation-Limited Reusable Rockets: A Quantum Reinforcement Learning Approach," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 8, pp. 12252-12257, August 2024.

[Access'24.08] [3] Hyunsoo Lee, **Gyu Seon Kim**, Minseok Choi, Hankyul Baek, and Soohyun Park, "Farthest Agent Selection with Episode-Wise Observations for Real-Time Multi-Agent Reinforcement Learning Applications," *IEEE Access*, vol. 12, pp. 108504-108514, August 2024.

◀ 2023 ▶

[ETRI'23.10] [2] **Gyu Seon Kim**, Haemin Lee, Soohyun Park, and Joongheon Kim, "Joint Frame Rate Adaptation and Object Recognition Model Selection for Stabilized Unmanned Aerial Vehicle Surveillance," *ETRI Journal*, vol. 45, no. 5, pp. 811-821, October 2023.

[TIV'23.08] [1] Chanyoung Park, **Gyu Seon Kim**, Soohyun Park Soyi Jung, and Joongheon Kim, "Multi-Agent Reinforcement Learning for Cooperative Air Transportation Services in City-Wide Autonomous Urban Air Mobility," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 8, pp. 4016-4030, August 2023.

Conferences

◀ Top-Tiers ▶

[GLOBEC'25] [13] Hyojun Ahn, Seungcheol Oh, **Gyu Seon Kim**, Soyi Jung, Soohyun Park, and Joongheon Kim, "Hallucination-Aware Generative Pretrained Transformer for Cooperative Aerial Mobility Control," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, Taipei, Taiwan, December 2025, pp. 5369-5374.

[WiOpt'25] [12] **Gyu Seon Kim**, Jaehyun Chung, Trung Q. Duong, Soohyun Park, and Joongheon Kim, "Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach," in *Proc. IEEE/IFIP International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt)*, Linköping, Sweden, May 2025, pp. 1-8.

[ICASSP'25] [11] **Gyu Seon Kim**, Samuel Yen-Chi Chen, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Coordinated Satellite Systems", in *Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Hyderabad, India, April 2025, pp. 1-5.

[WiOpt'24] [10] Sungjoon Lee, **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management," in *Proc. IEEE/IFIP International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt)*, Seoul, Republic of Korea, October 2024, pp. 305-312.

[ICC'23] [9] Chanyoung Park, Soohyun Park, **Gyu Seon Kim**, Soyi Jung, Jae-Hyun Kim, and Joongheon Kim, "Multi-Agent Deep Reinforcement Learning for Efficient Passenger Delivery in Urban Air Mobility," in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, June 2023, pp. 5689-5694.

◀ Others ▶

[ICEIC'27] [8] **Gyu Seon Kim**, Soohyun Park, Soyi Jung, and Joongheon Kim, "Learning-based Aircraft Taxi Routing for Traffic Management: Empirical Evaluation Study for Hartsfield-Jackson Atlanta International Airport," *Proc. IEEE International Conference on Electronics, Information, and Communication (ICEIC)*, Fukuoka, Japan, January 2027, pp. 00-00.

[PQAI'25] [7] **Gyu Seon Kim**, Samuel Yen-Chi Chen, Soohyun Park, and Joongheon Kim, "Satellite Constellation Scheduling with Quantum Reinforcement Learning", in *Proc. Springer Nature, Frontiers in Industrial Engineering. Purdue Quantum AI (Communications in Computer and Information Science, vol. 2724)*, West Lafayette, IN, USA, October 2025, pp. 51-56.

[ICTC'25] [6] **Gyu Seon Kim**, In-Sop Cho, and Joongheon Kim, "Strategic Role of LEO Satellites in the 6G Roadmap and Orbital Analysis," in *Proc. IEEE International Conference on Information and Communication Technology Convergence (ICTC)*, Jeju, Republic of Korea, October 2025, pp. 1996-2001.

[ICTC'24] [5] Hyunwoo Nam, **Gyu Seon Kim**, and Joongheon Kim, "Bicycle Rental Predictive Modeling with Machine Learning for Carbon Neutrality," in *Proc. IEEE International Conference on Information and Communication Technology Convergence (ICTC)*, Jeju, Republic of Korea, October 2024, pp. 2033-2038.

[ICTC'24] [4] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning: An Overview," in *Proc. IEEE International Conference on Information and Communication Technology Convergence (ICTC)*, Jeju, Republic of Korea, October 2024, pp. 1145-1150.

[APWCS'24] [3] Soohyun Park, **Gyu Seon Kim**, Soyi Jung, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning Software Design and Visual Simulations for Multi-Drone Mobility Control," in *Proc. IEEE VTS Asia Pacific Wireless Communications Symposium (APWCS)*, Singapore, August 2024, pp. 1-5.

[IJCAI'24] [2] **Gyu Seon Kim**, Yeryeong Cho, Soohyun Park, Soyi Jung, and Joongheon Kim, "Realizing Cooperative Global Internet Services for Space-Air-Ground Integrated Networks: A Quantum Multi-Agent Reinforcement Learning Framework," *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Quantum Algorithms, Optimization, and AI*, Jeju, Republic of Korea, August 2024.

[ICOIN'23] [1] **Gyu Seon Kim**, Hyunsoo Lee, Chanyoung Park, Soyi Jung, Jae-Hyun Kim and Joongheon Kim, "Hybrid MAC for Military UAV Networks," in *Proc. IEEE International Conference on Information Networking (ICOIN)*, Bangkok, Thailand, January 2023, pp. 656-659.

Publications (Domestic)

Journals (KCI)

◀ 2026 ▶

- [KIISE'26.10] [5] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based LEO Satellite Control and Communication," *Journal of Korean Institute of Information Scientists and Engineers (KIISE)*, (Accepted).
- [KIISE'26.02] [4] Sungjoon Lee, **Gyu Seon Kim**, Taejin Woo, and Soohyun Park, "Design of a Deep Reinforcement Learning Algorithm for Spatially Adaptive UAV Autonomous Navigation based on Transfer Learning," *Journal of Korean Institute of Information Scientists and Engineers (KIISE)*, vol. 53, no. 2, pp. 101-108, February 2026.

◀ 2024 ▶

- [KICS'24.03] [3] Jaehyun Chung, **Gyu Seon Kim**, Soohyun Park, Joongheon Kim, and W. Yun, "Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats," *Journal of Korean Institute of Communications and Information Sciences (KICS)*, vol. 49, no. 3, pp. 333-345, March 2024.

◀ 2023 ▶

- [KICS'23.09] [2] **Gyu Seon Kim**, Soohyun Park, Joongheon Kim, Yeong Goo Kim, Jaekyoung Ha, and Byung Hyun Jun, "Reinforcement Learning-Based Dynamic Routing for Robust Optimization of Low Earth Orbit (LEO) Satellite Communication Networks," *Journal of Korean Institute of Communications and Information Sciences (KICS)*, vol. 48, no. 9, pp. 1123-1134, September 2023.
- [KICS'23.02] [1] Chanyoung Park, **Gyu Seon Kim**, Kyeongjin Lee, and Ilsoo Yun, "Multi-Agent Deep Reinforcement Learning for Efficient Unattended Information Gathering and Monitoring of Autonomous UAM Systems," *Journal of Korean Institute of Communications and Information Sciences (KICS)*, vol. 48, no. 2, pp. 176-184, February 2023.

Conferences

◀ 2026 ▶

- [KICS'26] [17] **Gyu Seon Kim** and Joongheon Kim, "LLM-based Reinforcement Learning for UAV Control in Defense Systems," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2026, pp. 534-535.

◀ 2025 ▶

- [IEIE'25] [16] **Gyu Seon Kim** and Joongheon Kim, "Lightweight Control Modeling via Quantum Reinforcement Learning," in *Proc. Institute of Electronics and Information Engineers (IEIE) Conference*, Gyeongnam, Republic of Korea, November 2025, pp. 1584-1587.
- [KICS'25] [15] Jaehyun Chung, Hyojun Ahn, **Gyu Seon Kim**, and Joongheon Kim, "Recent Trends in Transformer Optimization for Improved Energy Efficiency," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Jeju, Republic of Korea, June 2025, pp. 574-575.
- [JCCI'25] [14] **Gyu Seon Kim**, Jaeyoung Choe, Soohyun Park, and Joongheon Kim, "Federated Reinforcement Learning-based Caching using Slimmable Neural Networks," *Joint Conference on Communications and Information (JCCI)*, Gangneung, Republic of Korea, April 2025.
- [KICS'25] [13] **Gyu Seon Kim**, Sungjoon Lee, Taejin Woo, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based Military UAV Control," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2025, pp. 291-292.
- [KICS'25] [12] Yun Seong Na, **Gyu Seon Kim**, and Joongheon Kim, "Optimal Algorithm Proposal for Urban Road Pathfinding Using Reinforcement Learning," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2025, pp. 520-521.

◀ 2024 ▶

- [KICS'24] [11] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based Self-Driving Vehicle Control for Carbon Neutrality," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Gyeongju, Republic of Korea, November 2024, pp. 207-208.
- [KICS'24] [10] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based Aircraft Longitudinal Static Stability Control," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Jeju, Republic of Korea, June 2024, pp. 1,094-1,095.
- [KICS'24] [9] **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based Lunar Module Landing Controller Design," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, January 2024, pp. 134-135.
- [KICS'24] [8] **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, and Joongheon Kim, "Slimmable Neural Networks Reinforcement Learning-based Robust Low Earth Orbit (LEO) Satellite Routing," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, January 2024, pp. 76-77.

◀ 2023 ▶

- [KAIC'23] [7] Emily Jimin Roh, **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, Soyi Jung, and Joongheon Kim, "Autonomous Torpedo Maneuvering and Waypoint Optimization via Reinforcement Learning," in *Proc. Korea Artificial Intelligence Conference (KAIC)*, Jeju, Republic of Korea, September 2023, pp. 177-178.
- [KICS'23] [6] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Utilizing Reinforcement Learning for Enhanced Efficiency in Fighter Airport Scheduling: A Case Study of Osan Air Base," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Jeju, Republic of Korea, June 2023, pp. 867-868.
- [JCCI'23] [5] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Low-Earth Orbit Satellite Routing based on Reinforcement Learning," *Joint Conference on Communications and Information (JCCI)*, Yeosu, Republic of Korea, May 2023.

- [JCCI'23] [4] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "A Mathematical Approach to Reinforcement Learning through the Bellman Equation," *Joint Conference on Communications and Information (JCCI)*, Yeosu, Republic of Korea, May 2023.
- [KICS'23] [3] Hyunsoo Lee, **Gyu Seon Kim** and Joongheon Kim, "Trends in Optimal Routing Technologies for Low Earth Orbit Satellite," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2023, pp. 406-407.
- [KICS'23] [2] **Gyu Seon Kim** and Joongheon Kim, "Communication & Sun-Pointing Mode Control Model For Cube Satellite," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2023, pp. 1088-1089.

◀ 2022 ▶

- [KICS'22] [1] **Gyu Seon Kim**, Hyunsoo Lee, Soyi Jung, J. H. Kim, and Joongheon Kim, "Self-Learning-Based Hybrid MAC for Military UAV Networks," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Gyeongju, Republic of Korea, November 2022, pp. 13-14.

Patents

United States

◀ 2025 ▶

- [U.S.'25.12] [9] Patent Application: No. 19/425,505, "Method and Apparatus for Routing Optimal Path in Low Earth Orbit Constellation Satellite Network System," December 2025.

Republic of Korea

◀ 2026 ▶

- [KR'26.07] [8] Patent Application: No. 10-2026-0133424, "Autonomous Unmanned Aerial Vehicle Mobility Control System and Method using Hierarchical Large Language Models and Safe Reinforcement Learning," July 2026.
- [KR'26.07] [7] Patent Application: No.10-2026-0128332, "Energy-Efficient Multi-Satellite Scheduling System and Method for Satellite Network using Initial Policy based on Large-Scale Language Model for Low-Earth Orbit Satellite Cluster," July 2026.

◀ 2025 ▶

- [KR'25.08] [6] Patent Application: No. 10-2025-0111715, "Multi-Agent Reinforcement Learning Based Anti-Drone Control System and Method," August 2025.
- * Patent Registration: No. 10-2991143, "Multi-Agent Reinforcement Learning Based Anti-Drone Control System and Method," July 2026. (Ministry of Intellectual Property, Republic of Korea)
- [KR'25.01] [5] Patent Application: No. 10-2025-0004523, "Multi-Agent Reinforcement Learning Device and Method," January 2025.

◀ 2024 ▶

- [KR'24.12] [4] Patent Application: No. 10-2024-0191045, "Ground Station and Method for Scheduling Non-Terrestrial Network Devices for Global Access in a Space-Air-Ground Integrated Network System," December 2024.
- [KR'24.12] [3] Patent Application: No. 10-2024-0199342, "Ground Station and Method for Cube Satellite Constellation Scheduling Based on Quantum Multi-Agent Reinforcement Learning," December 2024.
- [KR'24.12] [2] Patent Application: No. 10-2024-0199295, "Unmanned Aerial Vehicle Device Performing Communication in a Multi-Military Unmanned Aerial Vehicle Network System and Communication Method of the Unmanned Aerial Vehicle Device," December 2024.
- [KR'24.12] [1] Patent Application: No. 10-2024-0191045, "Method and Apparatus For Routing Optimal Path in Low Earth Orbit Constellation Satellite Network System," December 2024.

Review Experience

- IEEE Transactions on Aerospace and Electronic Systems
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Communications
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Network and Service Management
- IEEE Transactions on Cognitive Communications and Networking
- IEEE Transactions on Network Science and Engineering
- IEEE Transactions on Automation Science and Engineering
- IEEE Internet of Things Journal
- IEEE Communications Magazine
- IEEE Access
- IEEE/KICS Journal of Communications and Networks
- IEEE/KICS ICT Express
- Springer Nature Aerospace Systems

- ACM Computing Surveys
 - KSII Transactions on Internet and Information Systems
 - ETRI Journal
-

Talks

- **9. Workshop on Implementing Autonomous Vehicles using Modern Reinforcement Learning (01/2026, Seoul, Republic of Korea)**
– *Implementation of Multi-Agent Reinforcement Learning-based Unmanned Vehicle Control*
 - **8. Quantum Security Research Society Mid-Year Workshop (07/2025, Sejong, Republic of Korea)**
– *Theory and Applications of Quantum Reinforcement Learning*
 - **7. Understanding and Practice of Artificial Intelligence for Smart Mobility (03/2025, Seoul, Republic of Korea)**
– *Multi-Agent Reinforcement Learning-based Autonomous Urban Air Mobility Control for Aerial Transportation Services*
– *Multi-Agent Reinforcement Learning-based Advanced Air Traffic Management & Control*
 - **6. Workshop on Implementing Autonomous Vehicles using Modern Reinforcement Learning (01/2025, Seoul, Republic of Korea)**
– *Implementation of Multi-Agent Reinforcement Learning-based Unmanned Vehicle Control*
 - **5. A3 Foresight Program (06/2024, Jeju, Republic of Korea)**
– *Quantum Multi-Agent Reinforcement Learning for Joint Cube-Satellites and High-Altitude Long-Endurance Aerial Networks in SAGIN*
 - **4. Implementation and Simulation of Reinforcement Learning Algorithms Based on Unity (02/2024, Seoul, Republic of Korea)**
– *Application of Reinforcement Learning in Unity Environment Using ML-Agents – Practice in Various Scenarios*
 - **3. A3 Foresight Program (02/2023, Seoul, Republic of Korea)**
– *Communication & Sun-pointing Mode Control Model for Cube Satellite*
 - **2. Implementation and Simulation of Unity-Based Reinforcement Learning Algorithms for Telecommunication Networks (02/2023, Seoul, Republic of Korea)**
– *Application of Reinforcement Learning in Unity Environment Using ML-Agents – Practice in Various Scenarios*
 - **1. A3 Foresight Program (12/2022, Tokyo, Japan)**
– *Hybrid MAC for Military UAV Networks*
-

Teaching Assistants

- **SW Programming Basics (GECT002), Dept. of ECE, Korea University**
– *Spring 2025*
-

References

- **Prof. Joongheon Kim, Ph.D. Research and Dissertation Advisor**
– Professor, Korea University, Seoul, Republic of Korea
– E-mail: joongheon@korea.ac.kr
– Website: <https://sites.google.com/view/aimlab-kuee/members/joongheonkim>
- **Prof. Zhu Han, Ph.D. Visiting Research Supervisor**
– Professor, University of Houston, Houston, TX, USA
– E-mail: zhan2@uh.edu
– Website: <https://wireless.egr.uh.edu/>
– Website: <http://www2.egr.uh.edu/~zhan2/>